

Project Completion Report (PCR)

Project Name: Plutus Cheatsheet Generator

Project Number: 1400097

Project Manager: kwired

Project Start Date: December 2025

Project Completion Date: February 2026

1. Context & Original Approach

The Cardano ecosystem requires accessible, high-quality developer resources to lower the barrier to entry for building on Plutus. The Plutus Cheatsheet Generator was proposed in Fund 14 to address this by providing a comprehensive, community-driven platform for browsing, filtering, and debugging Plutus smart contracts. Our approach was to build a modern React-based web application with curated snippets, transitioning over time to include community contributions, advanced search features, and security-focused tutorials.

2. Deliverables & Outputs

We successfully delivered all proposed outputs across the three project milestones. The project is fully open-source under the MIT License.

2.1 Core Web Platform (Open Source)

- **Web App:** A high-performance React web application utilizing Vite and Tailwind CSS (glassmorphism design).
- **Snippet Library:** Over 66 smart contract snippets across Plutus V1, V2, and V3.
- **Export Functionality:** Users can export snippets seamlessly into Markdown or PDF formats for offline use or integration into their own repositories.
- **Source Code & License:** Hosted on GitHub under the MIT License.

2.2 Advanced Search & Versioning

- **Multi-Tag Filtering:** Users can filter snippets by complexity, use-case (DeFi, NFTs, Oracle, etc.), and Plutus Version (V1/V2/V3).
- **Version Tagging System:** Assures developers that the code they are looking at is compatible with their target Plutus version and the Conway era.

2.3 Interactive Tutorials & Security

- **Security Focus:** Created dedicated tutorials for common vulnerabilities such as Double Satisfaction, Replay Attacks, Datum Mismatch, and Minimum ADA Leakage.
- **Security Audit:** The platform and its code examples underwent a comprehensive security audit, resulting in zero critical issues. Examples reflect best practices for secure Cardano smart contract development.

2.4 Documentation Portal

- **Hosted Documentation:** A full documentation portal was developed and hosted on GitHub Pages (<https://kwired.github.io/Plutus-Cheatsheet-Generator/>).

- **Guides:** Includes setup instructions, user guides for platform features, and contribution guidelines.

2.5 Community Integration

- **Discord Bot:** Implemented a Discord bot to alert the community of new snippets and updates, fostering platform engagement.
 - **Community Snippets:** Successfully merged community-contributed snippets into the core repository.
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3. Usage & Impact

The Plutus Cheatsheet Generator has seen significant adoption by the Cardano developer community since its launch.

- **Active Users:** Analytics verify that the platform attracts over 1,000+ weekly active users.
 - **Developer Feedback:** The interactive tutorials and copy/export functionalities have significantly reduced the time developers spend searching for secure, reliable Plutus implementation patterns.
 - **Ecosystem Contribution:** By providing tested, audited, and versioned code snippets, the platform reduces the likelihood of vulnerable smart contracts being deployed to the Cardano mainnet.
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4. Evidence & Links

- **Live Web Application:** <https://plutus-cheatsheet.vercel.app/>
- **Documentation Portal:** <https://kwired.github.io/Plutus-Cheatsheet-Generator/>

- **GitHub Repository:** <https://github.com/kwired/Plutus-Cheatsheet-Generator>